

# William Jiao

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## EDUCATION

### University of California, Davis

*Bachelor of Science - Electrical Engineering*

Davis, California

Sep 2024 - Jun 2028

*Courses: Analog Circuit Design, Digital Circuit Design, Signals and Systems, Embedded Programming, Electricity and Magnetism*

## SKILLS SUMMARY

- **Design Tools:** Altium Designer, RapidHarness, LTSpice, STM32CubeIDE, KiCAD, ArduinoIDE, MATLAB, Excel
- **Programming Languages:** C, C++, Java, MATLAB, Python, Verilog
- **Tools:** GIT, UNIX Terminal, Oscilloscope, Power Supply, SD Formatter, CAN, SPI, I2C, SPICE Simulation

## TECHNICAL EXPERIENCE

### Formula Racing at UC Davis (FSAE)

Davis, CA

*Electrical Subteam Lead*

Sep 2024 - Present

- **Wiring Harness:** Lead design and assembly of a low and high voltage electric vehicle wiring harness
- **PCB Design:** Design PCBs to record live telemetry data in race car and handle critical high voltage safety faults
- **Real-Time Debugging:** Debug critical issues with wiring harness, PCBs, and high voltage system in real-time under high pressure and time constraints with electrical debugging tools
- **Microcontroller Programming:** Programming microcontrollers to control hardware timers and collect real-time data
- **Project Management:** Organize team members into projects, plan meetings with team members, create project timelines, and create project documentation
- **Sponsorship Outreach:** Reach out to sponsors to negotiate sponsorship deals, reduced electrical subteam costs by 50%

## PROJECTS

### Pipelined Matrix Multiplier

*Verilog, Quartus, ModelSim*

- Designed and implemented a pipelined matrix multiplier on an FPGA written with Verilog
- Analyzed timing waveforms in multiplier using ModelSim and pipelined data path to decrease cycle count
- Programmed FPGA using Quartus Prime, and debugged errors at compilation

### Telemetry Host - Ethernet PCB

*Altium Designer, CAN*

- Designed a high-speed Ethernet PCB in Altium Designer to stream live vehicle telemetry data to a router
- Routed differential pairs and optimized signal integrity for high-noise automotive environments
- Configured CAN bus interface to receive data transmitted from data acquisition hardware around the car
- Validated board power distribution networks and debugged hardware using oscilloscopes

### Telemetry Node - Data Acquisition PCB

*Altium Designer, STM32, CAN, LTSpice*

- Designed PCB to record live sensor data and transmit the data over CAN bus
- Implemented anti aliasing filter hardware and digital filtering in STM32 for sensor data
- Configured CAN bus interface to transmit processed telemetry data directly into telemetry host
- Optimized routing to prevent cross talk between CAN lines and to minimize size of PCB

### Cooling Testing Rig - Data Acquisition Hardware + Software

*Circuit Design, Arduino, STM32, SPI*

- Designed circuits to collect temperature and pressure readings from a testing rig
- Programmed Arduino to record and transmit data in real time to an SD card via SPI
- Programmed STM32 to variably control fan and pump speeds based on user inputs into test computer

### HVI - High Voltage Interface PCB

*Power Electronics, Altium, SPICE*

- Designed a hardware and software timer to activate safety circuitry to comply with FSAE rules
- Designed and validated current limiter hardware and discharge circuit using SPICE simulation
- Ensured galvanic isolation between high and low voltage drivers to maintain safety standards and FSAE rules

### Wiring Harness - LV Wiring Harness

*RapidHarness, Signal Integrity*

- Created wiring connections and wiring schematic of a low and high voltage wiring harness using RapidHarness
- Employed techniques to isolate high speed and low speed CAN bus data and power lines to prevent crosstalk
- Configured shutdown circuit in HV and LV wiring harness to comply with FSAE safety rules
- Organized design and assembly of the system's wiring harness to reduce harness assembly time by 3 months